# Mohammed Irshad C

Python Developer · Django, Backend & Full-Stack Systems

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## SUMMARY

Python developer who ships production systems end to end. Sole engineer on WorkshopOS, a 49-model Django / PostgreSQL platform (169,000 lines, 3,185 automated tests) replacing paper job cards, WhatsApp and spreadsheets at a premium car workshop, approved for go-live after staging by owners who had found no off-the-shelf software that fit. Two more products built solo and live: a real-time multiplayer game and a rehab app used daily through a real recovery. Ran a business before writing code, so cost, profit and cash are modelled as three separate figures.

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python · JavaScript (ES6+) · SQL · HTML5 · CSS3 |
| **Frameworks** | Django 5.2 · Django Channels (WebSockets, ASGI) · Django ORM · Bootstrap |
| **Databases** | PostgreSQL · SQLite · schema design · migrations · query optimisation (N+1 elimination, composite indexing) |
| **Frontend** | Vanilla JS · PWA · Service Workers · Web Push (VAPID) · Web Audio API · Web Speech API · mobile-first |
| **APIs & integrations** | REST · JSON APIs · LLM APIs (streaming, failover) · AWS SigV4 request signing · Cloudflare R2 (S3 API) |
| **Testing** | Django test framework · 3,500+ automated tests across projects · regression and invariant testing |
| **Infrastructure** | Railway · Render · PythonAnywhere · Cloudflare Workers · dev/prod settings · backup rotation · Git / GitHub |
| **Engineering** | System architecture · RBAC · decimal money logic · audit trails · async & concurrency · auth hardening |

## EXPERIENCE

**Software Developer (Freelance) · Formula D, Pullara, Kerala** 2026 – Present

*WorkshopOS "Titan": Django / PostgreSQL platform (private client repository)* Technical overview · System map

- Mapped how a fast-growing premium car workshop ran on paper job cards, WhatsApp and spreadsheets, then replaced all three with one system: job cards, inventory, supplier ledgers, payments, payroll, estimates, warranty claims, three-tier access.
- Delivered **49 models, 324 routes and 140 screens** in 294 commits (Jan – Oct 2026), backed by **3,185 automated Django tests** and 71 JavaScript tests: 1.34 lines of test for every line of application code.
- Built a **weighted-average inventory costing engine** as a date-ordered replay, chosen over FIFO so negative stock and back-dated supplier bills re-price correctly; per-batch cost is kept, so FIFO stays reconstructible.
- Isolated **21 view-free rule modules**, each owning one question (profit and cash, invoicing, settlement, money parsing), so no figure is derived two ways; money moves in **72 all-or-nothing transaction blocks** under a Change History audit trail.
- Hardened access with per-account and per-IP lockout, device-aware sessions with remote revocation and OTP reset; added Web Push (VAPID) and direct-to-storage photo uploads signed with stdlib AWS SigV4, verified against Amazon's test vector.

**Technical Consultant (Freelance) · Badusha Event Management** 2026

*Catering staff system (build pending client budget)* Specification · Original documents

- Specified a 10-role staff-management PWA to replace paper and WhatsApp coordination: **24 modules** ranked MVP, Optional or Deferred across six Django apps, written so other developers can build from it.
- Flagged India's DPDP Act risk for Aadhaar data and designed the retention notice to show at upload, never hidden.

## PROJECTS

**Word Hunt: Real-Time Multiplayer Game Engine** word-hunt-game.onrender.com

- **Django Channels / ASGI** with a six-state, server-authoritative state machine. A dropped socket marks a player disconnected, not removed, so a blip never costs their scored time bank; it reconnects with exponential backoff and full resync.
- Runs at **$0 a month**: an in-memory channel layer removes the Redis dependency, and every sound is synthesised with the Web Audio API, so there are no audio files. Server-side rate limiting and allowlisting on broadcast events; 104 tests.

**PhysioCoach: Offline-First Bilingual Rehab PWA** therapist.pythonanywhere.com

- Hands-free post-surgery knee rehab coach, built for a real recovery and **used daily through it**. A localStorage queue replays sessions after network loss and Screen Wake Lock keeps the display on, so the phone is never touched mid-routine; 80 tests.
- **Three-tier voice fallback** (Web Speech API, then 215 pre-generated English and Malayalam clips, then an on-demand server endpoint), with a stall timeout for the browser's known hung-utterance bug: reliable voice cues at no recurring API cost.

**Irshad's Agent: AI Assistant with Model Failover** irshaduu.github.io/portfolio

- The assistant on this portfolio: **five AI models in a failover chain** on a Cloudflare Worker, plus a job-post check; 136 tests.

## EDUCATION AND CERTIFICATIONS

**Python / Django Full-Stack Development** · Entri Elevate · Grade A+ (98%) · 2025
**Python Programming** · Certified User, Illinois Institute of Technology (08/2025)

## EARLIER BACKGROUND

**Founder, Event Management Company** (2019 – 2023). Built and scaled an events business, later handed to family: four years of P&L, staffing and supplier ownership, applied directly to the systems now being built.
**Algorithmic Trading** (2023 – 2024). Deployed and managed automated trading systems; hands-on risk management.